

CLERMONT-FERRAND AUVERGNE AP, France

WMO: 074600

Lat: 45.7869N

Lon: 3.1494E

Elev: 1090

StdP: 14.13

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.8	23.8	14.0	11.6	24.6	17.8	14.0	27.3	29.9	50.5	26.1	49.7	4.1	350	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	91.1	68.2	87.1	67.2	83.4	66.2	70.3	85.1	68.8	83.3	67.3	80.7	7.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.6	98.5	75.7	64.1	93.2	73.6	62.6	88.5	72.5	34.8	85.2	33.5	83.3	32.3	81.2	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	19.8	16.7	13.1	97.3	4.6	3.6	9.7	99.8	7.0	101.9	4.5	103.9	1.1	106.5		
(5)			12.3	73.6	4.5	1.8	9.1	74.9	6.4	75.9	3.9	76.9	0.6	78.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	39.6	41.0	46.7	51.3	58.2	64.9	68.7	68.4	61.5	55.6	45.9	40.8
	DBStd	12.56	8.14	8.11	7.01	6.26	6.43	6.90	6.02	6.50	6.21	7.44	7.41	8.01
	HDD50	1298	330	262	146	58	8	0	0	0	1	33	163	297
	HDD65	4580	787	673	567	413	226	86	30	35	140	299	574	750
	CDD50	2621	8	9	45	96	263	448	580	569	345	207	39	12
	CDD65	427	0	0	0	1	16	84	144	140	34	8	0	0
	CDH74	4449	0	0	2	29	221	888	1481	1438	339	51	0	0
	CDH80	1603	0	0	0	2	46	312	562	601	77	3	0	0
Wind	WSAvg	6.4	7.0	7.5	7.6	7.2	6.6	5.7	5.4	5.1	5.2	6.4	6.6	7.0
Precipitation	PrecAvg	23.4	1.3	1.1	1.1	1.7	3.5	2.7	1.9	2.9	2.3	2.1	1.5	1.4
	PrecMax	30.8	3.4	2.6	3.1	4.5	6.0	5.4	5.5	5.9	6.4	6.8	3.3	2.7
	PrecMin	17.5	0.2	0.2	0.0	0.2	0.7	0.2	0.3	0.5	0.2	0.2	0.4	0.1
	PrecStd	3.9	0.8	0.6	0.7	1.1	1.4	1.1	1.0	1.4	1.5	1.5	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.3	65.0	72.4	77.7	85.5	93.9	96.3	97.0	87.0	79.5	67.4	60.9
		MCWB	51.8	49.5	54.3	59.7	64.6	68.4	68.7	68.4	65.0	62.2	56.3	52.8
	2%	DB	57.1	59.6	67.2	72.7	79.9	87.8	90.3	91.3	81.9	74.7	63.5	57.8
		MCWB	50.2	49.0	52.0	56.9	62.8	67.0	68.4	68.1	64.3	61.4	55.1	51.0
	5%	DB	54.0	55.9	62.8	68.3	75.5	83.1	86.1	86.9	77.9	70.8	60.4	55.4
		MCWB	47.7	47.6	50.9	55.1	60.9	66.8	67.6	67.4	62.8	60.1	53.4	49.4
	10%	DB	51.4	53.1	58.9	63.7	71.3	79.1	82.2	82.2	73.7	67.3	57.2	52.7
		MCWB	46.0	46.0	49.5	52.9	59.2	65.4	66.1	66.0	61.3	58.7	51.6	47.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	52.1	56.0	61.3	67.2	72.2	72.3	71.7	69.0	64.8	58.0	53.7
		MCDB	59.1	60.7	68.4	74.7	79.8	85.5	86.9	88.0	80.0	75.2	65.3	59.9
	2%	WB	50.4	49.9	53.7	58.5	64.4	69.9	70.0	69.9	66.0	62.6	55.9	51.4
		MCDB	56.2	57.6	64.3	70.7	76.8	83.9	85.3	85.7	77.3	71.7	62.0	57.0
	5%	WB	48.2	48.2	51.7	55.7	62.0	67.9	68.4	68.3	64.1	60.8	53.9	49.7
		MCDB	53.7	55.2	61.4	65.8	72.9	80.6	83.0	83.4	75.0	68.7	59.4	54.8
	10%	WB	46.1	46.4	49.8	53.6	60.0	66.0	66.9	66.8	62.2	59.1	51.9	47.7
		MCDB	50.9	52.5	58.1	62.3	69.5	76.8	80.2	80.1	72.0	66.3	57.0	52.3
Mean Daily Temperature Range	5% DB	MDBR	12.7	15.2	18.6	19.3	20.1	20.8	21.7	21.9	20.5	17.6	14.2	12.8
		MCDBR	15.9	21.5	25.4	27.7	27.0	27.6	28.5	29.3	26.9	22.2	18.8	16.1
		MCWBR	10.7	13.4	14.2	14.9	13.5	12.5	11.9	11.7	12.4	11.7	11.8	10.9
	5% WB	MCDBR	14.6	19.1	22.4	24.7	24.2	24.8	25.7	25.8	23.1	20.2	16.7	14.9
		MCWBR	10.2	12.3	13.3	13.8	13.0	12.1	11.5	11.4	11.6	11.3	11.1	10.4
Clear-Sky Solar Irradiance	taub		0.295	0.314	0.344	0.373	0.379	0.387	0.382	0.377	0.351	0.345	0.312	0.293
	taud		2.506	2.453	2.396	2.321	2.349	2.358	2.364	2.389	2.442	2.469	2.512	2.512
	Ebn at Noon		260	275	281	281	282	279	279	276	275	260	251	249
	Edh at Noon		21	26	32	38	38	38	37	35	30	26	21	19
All-Sky Solar Radiation	RadAvg		399	669	1046	1393	1628	1906	1873	1641	1273	812	455	344
	RadStd		50	95	127	181	160	171	151	135	91	69	51	58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page